

SMOOTHED PARTICLE HYDRODYNAMICS THEORY AND PRACTICE

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The study of fluid flow problems is extremely important since almost everything on this planet moves with respect to fluid. In the human body, most somatic cavities and even most organs are filled by fluids. Those fluids interact with solid parts and so they participate in mechanical body characteristics by a large amount evidently. Ordinary computational packages can solve either deformable solid dynamics or fluid dynamics surrounded by rigid wall or they couple these two methods in iteration sense what means that fluid pressure acts on deformable wall which moves but this motion changes the fluid boundary and so the iteration loop closes until a small error in a time step. This article brings little different view on the stuff. Fluids are modeled by a gridless method called smoothed particle hydrodynamics.

Key words: hydrodynamics, simulation, particle, gridless method, Lagrangean approach

1. Smoothed particle hydrodynamics

There are two main approaches to model fluid flow. First one is *Eulerian approach* where an observer stays in a fixed space point watching the fluid flow around him. Second one is *Lagrangean approach* where the observer moves with the fluid. The Eulerian methods are called grid methods as well while the Lagrangean ones are called particle methods. *Smoothed particle hydrodynamics* is a Lagrangean method. The fluid is discretized into particles whose physical values change in sense of conservation laws. Physical property represents its average in the closest neighbourhood and is computed as a weighted sum over all nearest neighbours. The particles carry all computational information. They are themselves the computational framework on which the fluid equations are solved. The particles tend to repel each other and thereby keeps the computational framework regular. This makes the method easy to implement.

2. Fundamentals of the method

The base of the smoothed particle methods is an interpolation theory. Fluid mechanics conservation laws are transformed into integral equations using an interpolation function, so called *kernel*. The kernel refers to a weighting function and defines how much of each field variable contributes to the field variable at a given space point. It approximates δ -function. The kernel estimate of physical field value $A(\mathbf{r})$ is defined by

$$\langle A(\mathbf{r}) \rangle = \int_V A(\mathbf{r}') w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' , \quad (1)$$

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where $\mathbf{r}$ and $\mathbf{r}'$ are position vectors, V is a computational domain, h is so called *smoothing length* (see later) and $w(\mathbf{r} - \mathbf{r}', h)$ is a kernel with properties

$$\lim_{h \rightarrow 0} w(\mathbf{r} - \mathbf{r}', h) = \delta(\mathbf{r} - \mathbf{r}') \quad (2)$$

and

$$\int_V w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' = 1 . \quad (3)$$

If the physical field $A(\mathbf{r}')$ is known only on a set of N discrete points $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N$ with masses $m_1, m_2, \dots, m_N$ respectively, we can write down the relation

$$A(\mathbf{r}') = \sum_{j=1}^N A(\mathbf{r}_j) \delta(\mathbf{r}' - \mathbf{r}_j) (dV)_j \quad (4)$$

where $(dV)_j$ is a volume differential around spatial point $\mathbf{r}_j$. The second step is to express volume differential $(dV)_j$ as $\frac{m_j}{\varrho(\mathbf{r}_j)}$ where $\varrho(\mathbf{r}_j)$ is a density in point $\mathbf{r}_j$. Hence, approximation of value $\mathbf{A}(\mathbf{r})$ can be expressed as

$$\langle \mathbf{A}(\mathbf{r}) \rangle = \sum_{j=1}^N \frac{m_j}{\varrho(\mathbf{r}_j)} \mathbf{A}(\mathbf{r}_j) w(\mathbf{r} - \mathbf{r}_j, h) . \quad (5)$$

As an example, *smoothed* density can be computed as

$$\langle \varrho(\mathbf{r}) \rangle = \sum_{j=1}^N \frac{m_j}{\varrho(\mathbf{r}_j)} \varrho(\mathbf{r}_j) w(\mathbf{r} - \mathbf{r}_j, h) = \sum_{j=1}^N m_j w(\mathbf{r} - \mathbf{r}_j, h) . \quad (6)$$

In the second step, we construct relation for gradient $\mathbf{A}(\mathbf{r}')$. The procedure will be similar. Let us have equation

$$\langle \nabla_{\mathbf{r}'} \mathbf{A}(\mathbf{r}) \rangle = \int_V \nabla_{\mathbf{r}'} A(\mathbf{r}') w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' , \quad (7)$$

where $\nabla_{\mathbf{r}'}$ is a gradient with respect to primed coordinates. Integration by parts with

$$u = w(\mathbf{r} - \mathbf{r}', h) \Rightarrow du = \nabla_{\mathbf{r}'} w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' \quad (8)$$

and

$$dv = \nabla_{\mathbf{r}'} A(\mathbf{r}') d\mathbf{r}' \Rightarrow v = A(\mathbf{r}') \quad (9)$$

gives

$$\langle \nabla_{\mathbf{r}} \mathbf{A}(\mathbf{r}) \rangle = \mathbf{A}(\mathbf{r}') w(\mathbf{r} - \mathbf{r}', h)|_S - \int_V A(\mathbf{r}') \nabla_{\mathbf{r}'} w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' , \quad (10)$$

where S is the boundary of computational domain. Equation (10) has the gradient expressed with respect to the primed coordinates. We need to express it with respect to unprimed ones. In making this conversion we get a -1 out. One can show that

$$\nabla_{\mathbf{r}'} f = -\nabla_{\mathbf{r}} f . \quad (11)$$

With this relation and the assumption that the kernel, the variable or both go to zero at infinity, we can write equation (10) as

$$\langle \nabla A(\mathbf{r}) \rangle = \int_V A(\mathbf{r}') \nabla_{\mathbf{r}} w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}' . \quad (12)$$

The surface term does not necessarily vanish for finite systems. For these cases we must deal with boundary conditions explicitly. We are now ready to write down our smoothed gradient as the summation

$$\langle \nabla \mathbf{A}(\mathbf{r}) \rangle = \sum_{j=1}^N \frac{m_j}{\varrho(\mathbf{r}_j)} \mathbf{A}(\mathbf{r}_j) \nabla_{\mathbf{r}} w(\mathbf{r} - \mathbf{r}_j, h) . \quad (13)$$

The interpolating formulae are themselves typically valid to second order of the smoothing length h . This can be seen by the Taylor's expansion of $A(\mathbf{r}')$ around $\mathbf{r}' = \mathbf{r}$ as

$$\langle A(\mathbf{r}) \rangle = A(\mathbf{r}) + c h^2 \nabla^2 A(\mathbf{r}) + O(h^3)$$

in the integral interpolant where c is a constant independent on h . Equation (5) and (13) are considered as a heart of the smoothed particle hydrodynamics. From now, let us denote $A(\mathbf{r}_i) = A_i$, $\nabla = \nabla_{\mathbf{r}}$, $\mathbf{r} - \mathbf{r}' = \mathbf{r}_i - \mathbf{r}_j = \mathbf{r}_{ij}$ and $w(\mathbf{r} - \mathbf{r}', h) = w(\mathbf{r}_{ij}, h) = w_{ij}$.

3. Kernel and smoothing length

There is a variety of kernels possible to choose. Each type of kernel usually fits to another purpose simulated by smoothed particle hydrodynamics. One possibility is to use cubic function

$$w(\mathbf{r}, h) = \frac{\sigma}{h^n} \begin{cases} \frac{3}{2} \left(\frac{1}{2} z - 1 \right) z^2 + 1 , & 0 \leq z < 1 \\ \frac{1}{4} (2 - z)^3 , & 1 \leq z < 2 \\ 0 , & z \geq 2 \end{cases} , \quad z = \frac{|\mathbf{r}|}{h} , \quad (15)$$

where n is dimension of solved task and σ is a normalization constant. The advantage of the spline kernel is that has compact support and equals to zero for $|\mathbf{r}| > 2h$.

While the accuracy of numerical simulations by smoothed particle hydrodynamics depends weakly on the form of kernel function, the main factor that determines the accuracy is the width of the kernel function h which is comparable to the mean particle distance. Therefore, an increase in the number of particles in the calculation domain will decrease the error of smoothed particle hydrodynamics simulation. Naturally, the error depends on distribution of particles too. *Smoothing length* h represents effective width of the kernel. Effective width is a distance in which a particle can interact with its neighbours. This distance is equivalent to cell width in finite difference methods. Best results are obtained with 20–35 neighbours. One possibility to keep that number almost constant is to change smoothing length during computation. We can use relation

$$h_i = h_0 \left(\frac{\varrho_0}{\varrho_i} \right)^{\frac{1}{\nu}} , \quad (16)$$

where h_0 and ϱ_0 are initial values. It is necessary to keep same smoothing lengths for computation of forces from particle i to particle j and vice-versa. Otherwise third Newton's law would be attacked. Hence for computation of forces between particles i and j , we use average smoothing length

$$h_{ij} = \frac{h_i + h_j}{2} . \quad (17)$$

While the accuracy of numerical simulations by smoothed particle hydrodynamics depends weakly on the form of kernel function, the main factor that determines the accuracy is the width of the kernel function h which is comparable to the mean particle distance. Therefore, an increase in the number of particles in the calculation domain will decrease the error of smoothed particle hydrodynamics simulation.

4. Fluid flow

4.1. Momentum equation

The momentum equation for a non-viscous fluid in the absence of body forces is given by Euler's equation

$$\frac{d\mathbf{v}}{dt} = -\frac{1}{\varrho} \nabla p \quad (18)$$

in the material frame. $\mathbf{v}$ and p are velocity vector and pressure respectively. It can be computed directly that

$$\left\langle \frac{\nabla p}{\varrho} \right\rangle_i = \sum_{j=1}^N m_j \frac{p_j}{\varrho_i \varrho_j} \nabla w_{ij} \quad (19)$$

but the symmetrized form obtained from the identity

$$\frac{1}{\varrho} \nabla p = \nabla \left(\frac{p}{\varrho} \right) + \frac{p}{\varrho^2} \nabla \varrho \quad (20)$$

leads to exact linear and angular momentum conservation since it leads to the anti-symmetric inner forces. Hence, the momentum equation in the material frame can be written as

$$\frac{d\mathbf{v}_i}{dt} = - \sum_{j=0}^N m_j \left(\frac{p_j}{\varrho_j^2} + \frac{p_i}{\varrho_i^2} \right) \nabla w_{ij} . \quad (21)$$

4.2. Continuity equation

Density on each particle can be computed using relation (6) but it considers one more cycle per all particles during computation. Overmore, relation (6) causes density decrease by the boundary. Ghost particles symmetrically set beyond the boundary can solve that problem but smarter solution is to rewrite continuity equation

$$\frac{d\varrho}{dt} = -\varrho \nabla \cdot \mathbf{v} \quad (22)$$

to the smoothed form according to

$$\frac{p}{\varrho} \nabla \cdot \mathbf{v} = \frac{p}{\varrho^2} [\nabla \cdot (\varrho \mathbf{v}) - \mathbf{v} \cdot \nabla \varrho] \quad (23)$$

as

$$\left\langle \frac{d\rho}{dt} \right\rangle_i = \sum_{j=1}^N m_j (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla w_{ij} \quad (24)$$

and integrate it together with the equation of motion.

4.3. Heat transfer

Considering compressible flow, we have to take temperature changes into account. Energetical equation in the material frame

$$\frac{du}{dt} = -\frac{p}{\rho} \nabla \cdot \mathbf{v} \quad (25)$$

where u denotes internal energy can be expressed using relation (23) as

$$\left\langle \frac{du}{dt} \right\rangle_i = \sum_{j=1}^N m_j \frac{p_j}{\rho_j^2} (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla w_{ij} \quad (26)$$

in the smoothed form.

4.4. State equation

State equation ties together all equations describing conservation laws in fluid dynamics. That is why that equation must be chosen to represent appropriate flow and its compressibility or incompressibility. For perfect gas, we can write down

$$p = \rho R T, \quad (28)$$

where T absolute temperature and because

$$R = c_p - c_v, \quad (28)$$

where R is universal gas constant, c_p and c_v are specific thermal capacities by constant pressure and volume respectively, we obtain

$$p = \rho (\kappa - 1) u \quad (29)$$

since $u = c_v T$ and $\kappa = c_p / c_v$. It also implies that the sound speed can be computed as

$$c = \sqrt{(\kappa - 1) u}. \quad (30)$$

To rewrite equations (29), (30) and (31), we just add subscript i to variables p , ρ , T , u and c . Let us consider an incompressible fluid now. In grid methods, incompressibility can be expressed purely by constant density. In smoothed particles method, pressure is an explicit function of density and particles flow according to the density gradient. Hence we have to allow density variations. The artificial equation of state for slightly compressible fluids has been developed in form

$$\Delta p = K \frac{\Delta \rho}{\rho_0}. \quad (31)$$

However, casual liquids such as water are better describe by strongly nonlinear dependance of Δp on $\Delta \varrho$. The nonlinearity is added by a power to equation (31) which overcomes to

$$p = p_0 + K \left[\left(\frac{\varrho}{\varrho_0} \right)^\gamma - 1 \right], \quad (32)$$

where p_0 and ϱ_0 are initial values of pressure and density respectively. Further parameter γ introduced here tasks is a constant usually set to 7. For incompressible fluids such as water, a prohibitively small time step must be taken if we used the actual equation of state because of Courant-Friedrich-Levy condition. Because of this, artificial equations of state have been defined for situations where the speed of sound is low enough to be practical yet and high enough to limit density fluctuations. Hence, in smoothed particle hydrodynamics, bulk modulus K is the real bulk modulus or it can be expressed in form

$$K \geq 100 \varrho_0 \frac{v_{max}^2}{\gamma}, \quad (33)$$

where v_{max} is maximum velocity in the domain. Bulk modulus expressed in form (33) fits better to smoothed particle hydrodynamics purposes. Equation (33) implies that sound speed can be written as

$$c = 10 v_{max} \left(\frac{\varrho}{\varrho_0} \right)^{\gamma-1} \approx 10 v_{max}. \quad (34)$$

Finally, equation (32) can be rewritten to the smoothed particle hydrodynamics form as

$$p_i = p_0 + K \left[\left(\frac{\varrho_i}{\varrho_0} \right)^\gamma - 1 \right] \quad (35)$$

for slightly compressible fluids.

5. Artificial viscosity

From now, let us consider the equality

$$\langle A \rangle_i = A_i \quad (36)$$

for any field A . Equations derived uptill now describe non-viscous flows. If we want to describe viscous fluid, we have to introduce a mechanism which will convert kinetic energy into heat. Generally, term

$$\frac{d\mathbf{v}_i}{dt} = - \sum_{j=1}^N m_j \left(\frac{p_i}{\varrho_i^2} + \frac{p_j}{\varrho_j^2} + \Pi_{ij} \right) \nabla w(\mathbf{r}_{ij}, h) \quad (37)$$

is added into momentum equation where Π_{ij} is defined as a *viscous pressure* and is given by

$$\Pi_{ij} = \frac{1}{\varrho_{ij}} (-\alpha \mu_{ij} c_{ij} + \beta \mu_{ij}^2) \quad (38)$$

and

$$\mu_{ij} = \begin{cases} h_{ij} \frac{\mathbf{v}_{ij} \cdot \mathbf{r}_{ij}}{|\mathbf{r}_{ij}|^2 + h_{ij}^2 \eta^2}, & \mathbf{v}_{ij} \cdot \mathbf{r}_{ij} < 0, \\ 0, & \mathbf{v}_{ij} \cdot \mathbf{r}_{ij} \geq 0, \end{cases} \quad (39)$$

where α , β and $\eta^2 = 0.01h^2$ are constants. Another approach arises from direct approximation of the viscous term as

$$\left\{ \left(\frac{1}{\varrho} \nabla \cdot \mu \nabla \right) \mathbf{v} \right\}_i \approx \sum_{j=1}^N \frac{m_j (\mu_i + \mu_j) \mathbf{r}_{ij} \cdot \nabla w_{ij}}{\varrho_i \varrho_j (\mathbf{r}_{ij}^2 + 0.01 h^2)} \mathbf{v}_{ij} , \quad (40)$$

where $\mu = \varrho \nu$ is dynamic viscosity and momentum equation can be rewritten into

$$\frac{d\mathbf{v}_i}{dt} = - \sum_{j=1}^N m_j \left(\frac{p_i}{\varrho_i^2} + \frac{p_j}{\varrho_j^2} \right) \nabla w(\mathbf{r}_{ij}, h) + \sum_{j=1}^N \frac{m_j (\mu_i + \mu_j) \mathbf{r}_{ij} \cdot \nabla w_{ij}}{\varrho_i \varrho_j (\mathbf{r}_{ij}^2 + 0.01 h^2)} \mathbf{v}_{ij} + \mathbf{F}_i , \quad (41)$$

where $\mathbf{F}_i$ is a body force per unit mass acting on particle i .

6. Dynamic pressure

For low Reynold's numbers, local variations in pressure gradient which force fluid motion can be very small in comparison to the hydrostatic pressure gradient. This is of special significance to smoothed particle hydrodynamics method since pressure is obtained using an explicit function of density. Consequently for many problems, it is simpler to model the *dynamic pressure* defined as

$$p_d = p - p_0 , \quad (42)$$

where p and p_0 are the total and hydrostatic pressures respectively. Since pressure appears in the Navier-Stoke's equation only as an gradient, the effect of hydrostatic pressure is to the body force. We can write down

$$-\frac{1}{\varrho} p = -\frac{1}{\varrho} p_d - \frac{1}{\varrho} p_h . \quad (43)$$

Substituting relation (43) into equation the Navier-Stoke's equation

$$\frac{d\mathbf{v}}{dt} = -\frac{1}{\varrho} p + \nu (\nabla \cdot \nabla) \mathbf{v} + \mathbf{g} \quad (44)$$

gives

$$\frac{d\mathbf{v}}{dt} = -\frac{1}{\varrho} p_d + \nu (\nabla \cdot \nabla) \mathbf{v} + \mathbf{g} - \frac{1}{\varrho} p_h = -\frac{1}{\varrho} p_d + \nu (\nabla \cdot \nabla) \mathbf{v} + \mathbf{F} , \quad (45)$$

where $\mathbf{F}$ is the body force defined as

$$\mathbf{F} = \mathbf{g} - \frac{1}{\varrho} p_h . \quad (46)$$

Even the state equation (32) can be simplified to

$$p_d = K \left[\left(\frac{\varrho}{\varrho_0} \right)^\gamma - 1 \right] . \quad (47)$$

For simplicity, p is used to denote p_d in those sections. Also the dynamic pressure is modelled by equations (21) and (41).

7. Searching for nearest neighbours

In contrast to a grid-based method where the locations of neighbouring grid-cells are found directly per definition of the grid, smoothed particle hydrodynamics is dependent on fast techniques for finding the neighboring particles that contributes to the smoothed particle hydrodynamics summations. Otherwise, the method will degrade into a direct summation where computational time is scaled by N^2 . Since the forces between any couple of particles are antisymmetric, we can simply decrease the computational time to be scaled by $N(N - 1)/2$. For one-dimensional problem, it is simple to reach the computational time scaled by much less than N . At each time step, we sort the particles according to their coordinates. Hence we can detect nearest neighbours just by their indices.

8. Initial setup and relaxation

In a general smoothed particle hydrodynamics calculation, one needs to initially specify particle masses, positions, velocities and in some cases temperatures or internal energies. All of these except the positions are usually straight forward to specify according to the initial conditions. One thing to keep in mind is that the initial conditions must not exceed the available resolution, thus there should not be sharp discontinuities. There are many ways to distribute particle positions but the method of choice depends to a high degree on the problem at hand. For relatively homogeneous systems it is often convenient to place particles on a cubic grid. A body-centred lattice have an advantage for such cases since it gives more close neighbours.

One of the main advantages of smoothed particle hydrodynamics is its ability to arrange its own computational grid. This is because the kernel function is centrally peaked and therefore particles tend to repel each other and maximise their inter-particle distances and thereby achieving a dynamical equilibrium. This is also the ideal particle distribution to start an smoothed particle hydrodynamics simulation with. There are several ways to relax an smoothed particle hydrodynamics system depending on the situation. A static stable equilibrium can be relaxed using a large artificial viscosity.

9. Error estimation

This have remained the difficult part of smoothed particle hydrodynamics. Although it is possible to see how different errors arise in smoothed particle hydrodynamics, it is difficult to make analytical estimates. It is possible to make analytical estimates of statistical errors but these generally require the particle distribution to be known. Early estimates assumed the particle distribution to be random which resulted in correspondingly large statistical errors typical for random distribution Monte-Carlo integrations. In practise, particle distributions is or rather should be in dynamical equilibrium, thus relaxed, and the distribution is not random, therefore the measured statistical errors were typically much smaller than expected.

There are generally three types of systematic errors. Two are again due to the particle distribution and the kernel always being centred on the particle. The third error can be seen from equation (14), when the represented field has a non-vanishing Laplacian. This error is also dependent on the spatial resolution. Therefore, direct tests and comparison with known solutions are often better suited. These should be selected depending on the problem at hand since even fairly simple tests can be time consuming to implement.

A relatively simple way to check for sampling errors due to particle disorder or errors in the implementation is to investigate a unit field. When putting $f(\mathbf{r}_j) = 1$ in equation (5), the result should ideally sum up to unity and the gradient equation (7) to zero.

There are several other useful diagnostic checks. Globally conserved quantities as total energy, linear and angular momentum, total entropy can be checked. Initial sudden jumps may indicate a need for relaxation. Individual particles are also good to check for suspicious values. The importance of having a good plotting and analysis tool must be emphasised. Without plotting various quantities it is difficult to even detect errors and problems.

10. Modelling of the boundary

If the surface term in relation (12) does not go to zero within our domain, we have to apply explicit boundary conditions. The boundary of the domain is modelled using finite element method in two and three dimensions. The case of one dimension, which can be discussed as well, has no meaning for boundary modelling. In two dimension, beam finite element model is derived. To model thin boundaries in two dimensions, beam element theory is introduced. Since the usage in the human body where the flexural stiffness is neglectible in most cases, the conversion of the beam finite element into a bar finite element is derived as well. For thick boundary, full two-dimensional element consisting of three or four nodes is presented. Three dimensions are represented by simple linear elastic tetrahedron element. Since the theory for presented finite elements is known, it is not presented here.

11. Coupling between smoothed particles and finite elements

Both free-slip and no-slip conditions which can be simply applied in finite difference methods per grid definition can not be directly used since the particles move. Especially for human tissues whose undergo large deformation, special contact between the flow and the boundary has to be developed. From my experience in dealing with the contact problem in computational solid dynamics, the particle-to-element penalty contact method is a good starting point.

11.1. Contact with the boundary

Generally, we define so called contact thickness which is an area with constant thickness along each element. For each couple of particles and elements, we detect if the particle enters that area. If yes, penetration p which is a deep of particle's penetration into the contact thickness is computed. The particle undergoes a penalty force normal to the boundary element which depends on the deep of penetration and acts in the boundary normal direction against the penetration. All the contacts act to the particle by some force. Obeying the third Newton's law, the oposite force must act to the element. The place of acting is in the particle projection to the element.

12. Time integration

Motion of both particles and nodes is driven by their accelerations. Accelerations of particles are given by equation (41). Motion of finite element nodes is driven by equation

$$\mathbf{M} \ddot{\mathbf{q}} + \mathbf{B} \dot{\mathbf{q}} + \mathbf{K} \mathbf{q} = \mathbf{Q} . \quad (48)$$

Since matrix $\mathbf{M}$ is positively definite, we can use fast conjugate gradient method to solve that system of equations. Now, the motion of the system can be integrated by standard rules. I have used the centered acceleration method for motion of both particles and elements. Sometime, for obtaining better results, the so called X -variant of the smoothed particle hydrodynamics is used. It updates the velocities of smoothed particles as

$$\frac{d\mathbf{x}_i}{dt} = \hat{\mathbf{v}}_i = \mathbf{v}_i + \varepsilon \sum_{j=1}^N m_j \frac{\mathbf{v}_{ji}}{\varrho_{ij}} w_{ij} , \quad (49)$$

where $0 \leq \varepsilon \leq 1$ is a constant. The reason is that the particles move with a velocity that is closer to the average velocity in the neighbourhood and keeping the particles in more order. Since the used scheme is explicit, the time step has to fulfill convergence stability conditions

$$\Delta t = k_{stab} \min_k k \frac{l_k}{c_k} \quad (50)$$

determined by local speed of sound. Here l_k denotes local length. Condition (50) implies

$$\Delta t \leq c_{stab} \min_{i \in \{1, \dots, \text{number_of_particles}\}} \left\{ \frac{h_i}{c_i + 0.6 (\alpha c_i + \beta \mu_i)} \right\} , \quad (51)$$

$$\Delta t \leq k_{stab} \min_{j \in \{1, \dots, \text{number_of_elements}\}} \left\{ \frac{L_j}{c_j} \right\} , \quad (52)$$

where relation (51) comes from the Courant-Friedrich-Lewy's condition considering artificial viscosity term where Courant's number c_{stab} should be taken around 0.3 up to 0.9 and relation (52) arises from the finite element theory [15]. L_i is the ratio of the i^{th} element volume to large side area. In case of beams or bars, L_i is the length of the i^{th} element. k_{stab} is a time step factor usually chosen less than one.

13. Solver

I have merged all the knowledges into a computational package called *SPHCOFEM*. The *MATLAB* and *C* languages were used. *MATLAB* serves only for pre-processing and post-processing since graphical operations are very simple to program. The solver is programmed in *C* since it is very fast comparing to *MATLAB*. Overmore, using the supercomputer at the University of West Bohemia supercomputer center, the computations can be provided in reasonable time. Brief algorithm of the solver can be expressed as

1. reading initial data
2. WHILE actual time $\leq$ termination time
3. update time step
4. move everything to half of the time step
5. compute accelerations, changes of density and changes of internal energy
6. – compute contact forces
7. – compute particle accelerations
8. – compute nodal accelerations
9. move everything to end of the time step
10. update actual time
11. END WHILE.

14. Validation of the code

Since I have not provided any experiments for comparison of results, the validation tasks are fully based on analytical or experimental results, both found or measured in literature.

15. Shock-tube

This validation sources fully from [9] where analytical solution can be obtained. Firstly, let us describe the problem. The measurement can be provided in laboratory with a stationary gas divided by a into two regions, each characterised by a given pressure and density as shown in Figure 1.

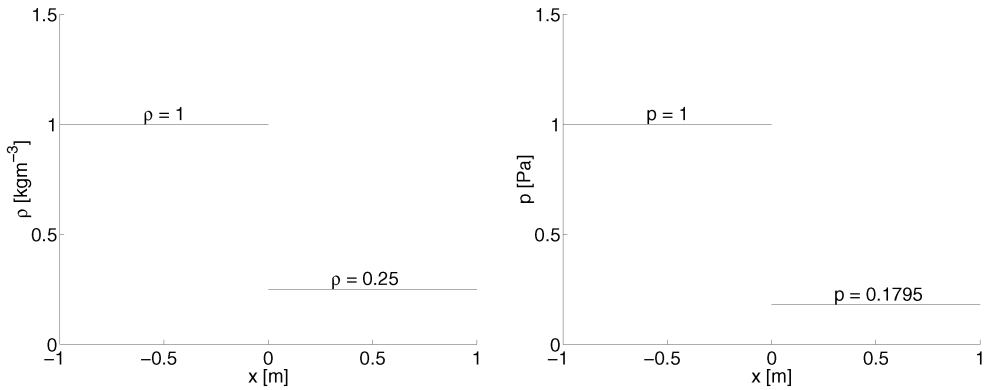


Fig.1: Shock-tube set up

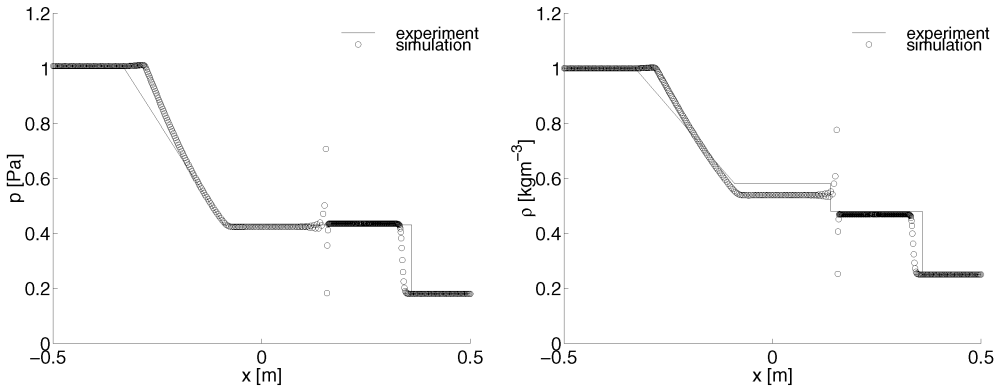


Fig.2: Shock-tube results

At the beginning, the diaphragm is broken. From that moment, a refraction wave starts travelling to the left into the region of higher density and both contact discontinuity and shock front travel to the right. As was said previously, the problem can be either measured or solved analytically. Figure 2 compares *SPHCOFEM* code to results obtained by analytical solution.

For simulation by the *SPHCOFEM* code, 1000 particles were used. Artificial viscosity coefficients α and β were set both to unity, specific heat ratio $\kappa = 1.4$. Simulation time

is 0.230768 s . Black solid line shows analytical data whilst small blue circles presents the numerical results.

The influence of absence of the boundary is not important in the shock-tube problem since long before the rarefaction waves, which start travelling inwards from the boundaries, reach the area where the relevant things happen, the features we want to see have already developed.

One can see the discontinuity in both pressure and density during the simulation. They are probably caused by the shock which was even not smoothed by the method. A possible solution could be to tune another artificial viscosity parameters or even the artificial viscosity relation which sketches tasks further development.

15.1. Breaking dam

This simulation shows the availability to use smoothed particle hydrodynamics for free surface flow problems. A rectangular column of water in hydrostatic equilibrium is confined between the vertical wall. The water column is 0.05715 m high and 0.05715 m wide. At time $t = 0$, the gate is removed instantaneously and the water is allowed to collapse under the gravity force.

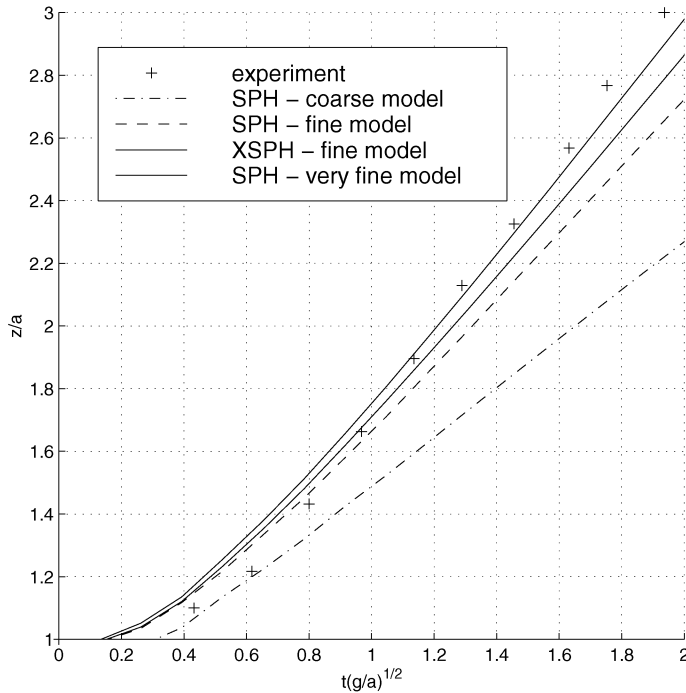


Fig.3: Surge position comparison to experimental data

The mathematical model is represented by 3249 particles. Figure 3 shows the simulation results against experimental data. For convenience, all used variables are dimensionless by choosing the initial width of dam a as a reference length. Hence, dimensionless time and surge front position are defined as $t\sqrt{g/a}$ and $\frac{z}{a}$ respectively where $g = -9.81\text{ ms}^{-2}$ is the gravitational acceleration and z is the surge front. The water has density $\rho = 996\text{ kgm}^{-3}$.

Coefficients α and β are chosen 1.2 and 1.5 respectively. Soft contact between the water and the boundary without any friction is defined.

The pluses shows the experimental data whilst dashed blue line shows numerical results. Furthermore, the influence of number of particles and the X -variant is presented. Green dash-dotted line shows inaccurate results using only 361 particles. On the contrary, red solid line presents more accurate results obtained by the X -variant of the method with parameter $\varepsilon = 0.1$. However, that variant is much more time consuming. Very large number of particle (12996) presents the solid black line which is the most exact solution here.

Standard smoothed particle hydrodynamics as well as X -variant computational results are presented. Figure 4 presents chosen states of simulation.

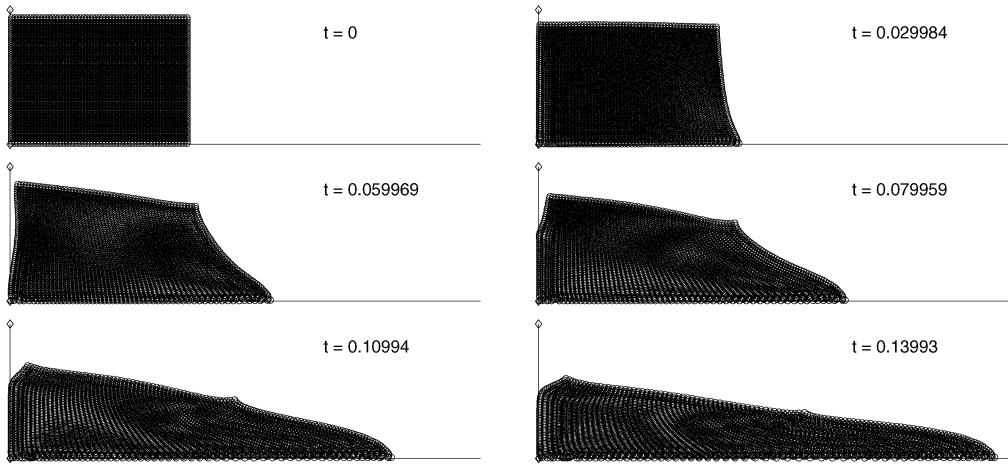


Fig.4: Breaking dam

16. Applications

Last section presents various tasks which might be solved using smoothed particle hydrodynamics method. Unfortunately, the task shown has no experimental comparison. It deals with curved deformable tube with flowing water inside. Figure 5 shows the numerical results of the interaction.

The water is initiated by initial velocity $v_0 = 1 \text{ ms}^{-1}$ for all particles. There is a sliding contact interface between the particles and the boundary defined. The flowing particles interact with the deformable boundary during their motion. Unfortunately, the definition of sliding contact interface do not allow to develop viscous parabolic surge (to be compared for example to Poiseuille's flow) since the particles are moving even along the wall. Possible solution is to tune contact friction characteristics which is another task for further development. Also in planned development, there is to simulate solid boundary using smoothed particles as well which will avoid the interaction problems.

17. Conclusion

Computational code solving the problem of fluid flow and its interaction with deformable boundaries was developed. The core of the package is a Lagrangean method for fluid flow simulation called smoothed particle hydrodynamics.

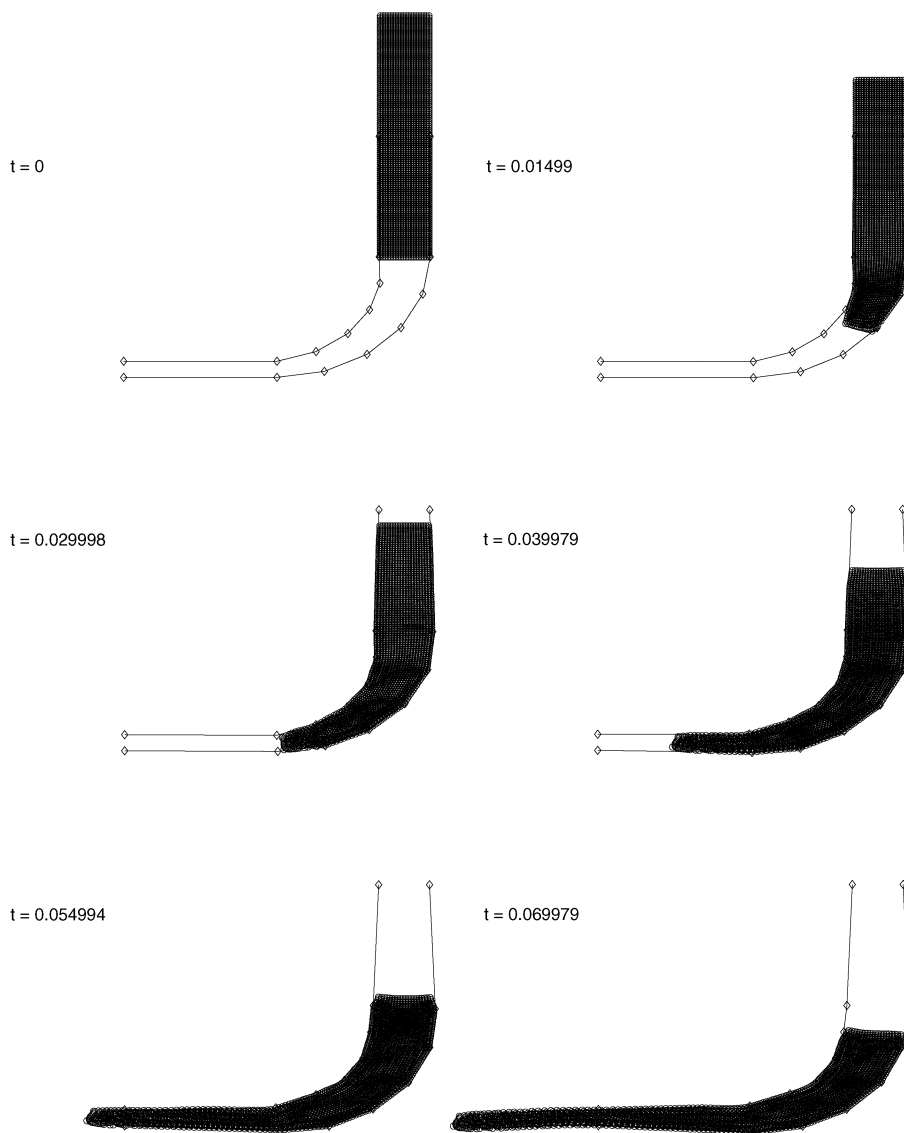


Fig.5: Fluid flow within deformable boundary

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References

- [1] Attaway S.W., Heinstein M.W., Swegle J.W.: Coupling of Smoothed Particle Hydrodynamics with the Finite Element Method, Nuclear Engineering and Design 150, pp.199–205, 1994
- [2] Bonet J., Kulasegaram S., Lok T.-S.L.: Corrected Smoothed Particle Hydrodynamics Methods for Fluid and Solid Mechanics Computation, European Conference on Computational Mechanics '99, Munich, 1999

- [3] Bonet J., Lok T.-S.L.: Variational and momentum preservation aspects of Smoothed Particle Hydrodynamic formulations, *Computed Methods Apl. Mech. Engrg.* 180, pp. 97–115, 1999
- [4] Cimrman R.: Metoda řešení Burgesovy rovnice, diploma thesis, ZČU, 1998
- [5] Dupal J.: Výpočtové metody mechaniky, undergraduate text, ZČU, 1998
- [6] Fagan M.J.: Finite Element Analysis – Theory and Practice, Longman, 1992, ISBN 0-582-02247-9
- [7] Fulk D.A., Quinn D.W: An Analysis of 1-D Smoothed Particle Hydrodynamics Kernels, *Journal of computational physics* 126, pp. 165–180, 1996
- [8] Gingold R.A., Monaghan J.J: Kernel Estimates as a Basis for General Particle Method in Hydrodynamics, *Journal of computational physics* 46, pp. 429–453, 1982
- [9] Henneken E.A.C.: Smoothed Particle Hydrodynamics, doctoraal-scriptie, Sterrewacht, Leiden
- [10] <http://www.astro.uu.se/~pohlman/SPH/SPH-course.html>
- [11] Le Veque R.J., Mihalas D., Dorfi E.A., Müller E.: Computational Methods for Astrophysical Fluid Flow, Saas-Fee Advanced Course 27, Lecture Notes, Swiss Society for Astrophysics and Astronomy, 1997
- [12] Monaghan J.J.: An Introduction to SPH, In.: *Computer physics communications* 48, pp. 86–96, North Amsterdam, 1998
- [13] Monaghan J.J.: Smoothed Particle Hydrodynamics, *Annual Reviews Astron. Astrophys.*, pp. 543–574, 1992
- [14] Nelson R.P., Papaloizou C.B.: Variable Smoothing Length and Energy Conservation in Smoothed Particle Hydrodynamics, *Mon. Not. R. Astron.*, pp. 1–20, 1994
- [15] PAM-CRASH/SAFE™/PAM-SAFE™ Reference And Solver Notes Manuals, Version 2001
- [16] Pozorski J., Wawreńczuk A.: Smoothed Particle Hydrodynamics Modelling and Computation of Incompressible Viscous Flows, *Colloquium Fluid Dynamics '99*, Prague, 1999
- [17] Stellingwerf R.F.: Smooth Particle Hydrodynamics, Mission Research Corporation, Albuquerque, NM USA
- [18] Takeda H., Miyama S.M., Sekiya M.: Numerical simulations of viscous flow by smoothed particle hydrodynamics, In.: *Prograss of theoretical physics*, Vol. 92, No. 5, November 1994
- [19] Welton W.C.: Two-Dimensional PDF/SPH Simulations of Compressible Turbulent Flows, *Journal of computational physics* 139, pp. 410–443, 1998

Hydrodynamika vyhlazených částic – teorie a praxe

Klíčová slova : *hydrodynamika, simulace, částice, Lagrangeovský přístup*

Práce se zabývá popisem proudění tekutin metodou vyhlazených částic. Metoda vyhlazených částic je Lagrangeovská metoda, kde pohybující se částice nesou veškerou informaci. Hodnota jakékoliv mechanické veličiny na částici se počítá jako vážený průměr přes všechny sousedící částice. Metoda vychází z poznatků teorie aproximace. Integrální vztahy se po diskretizaci přepíší pomocí sum. Těmito vztahy se potom upraví všechny rovnice popisující proudění tekutin. Hranice je ve vytvořeném programu modelována pomocí jednoduchých konečných prvků. Ty interagují s vyhlazenými částicemi pomocí kontaktního algoritmu na bázi penetračních sil. Metoda je vhodná hlavně pro úlohy s velkými přesuny hmoty či velkými deformacemi hranice, čehož lze využít hlavně v biomechanice.

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